

Smart **Agile** Resource Optimizer

A Hybrid Predictive & Prescriptive AI Framework for Mission-Critical Sprint
Planning and Task Allocation

The Planning Paradox

Traditional Resource Management is Breaking in Complex Environments

INEFFICIENCIES IN MANUAL PLANNING

Estimating Effort

Human "gut-feel" estimates are consistently prone to optimism bias, leading to over-commitment and sprint spills across the lifecycle.

Allocation Waste

Junior talent is often overwhelmed by complexity, while senior specialists are underutilized on low-priority "easy" tickets.

THE HYBRID ARCHITECTURE



ML Predictor

Gradient Boosting Regressors analyze historical Jira data to predict task duration with scientific accuracy.



Genetic Optimizer

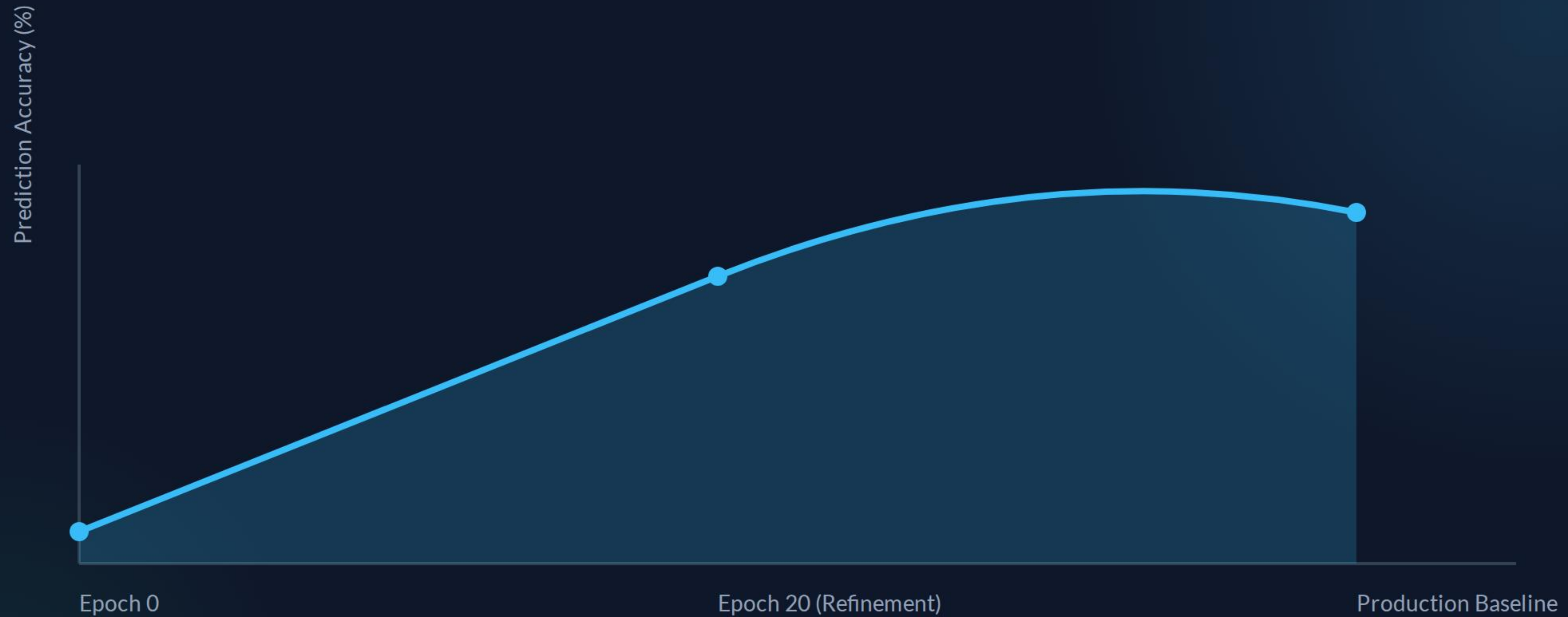
Simulated evolution explores millions of schedule permutations to find the lowest-cost, highest-speed path.



Constraint Solver

Real-world rules (skills, workload caps, and dependencies) are enforced as mathematical fitness penalties.

PREDICTIVE ML ENGINE ACCURACY



Leveraging Gradient Boosting to achieve a baseline Mean Absolute Error (MAE) of ~0.25 days per task.

SYSTEMIC IMPACT METRICS

90%

Planning Time Reduction

Automated Intelligence

By automating the combinatorial math of resource leveling, project managers shift from manual "spreadsheet shuffling" to strategic decision support.

The Genetic Algorithm iterates through thousands of generations in under 2 seconds to deliver a near-optimal sprint plan.

RESOURCE CAPABILITY MATRIX

Profile Level	Max Complexity	Cost per Day	Workload Threshold
Senior Developer	Level 3 (Hard)	\$600.00	5.0 Days / Sprint
Mid-Level Developer	Level 2 (Medium)	\$300.00	5.0 Days / Sprint
Junior Developer	Level 1 (Easy)	\$150.00	5.0 Days / Sprint

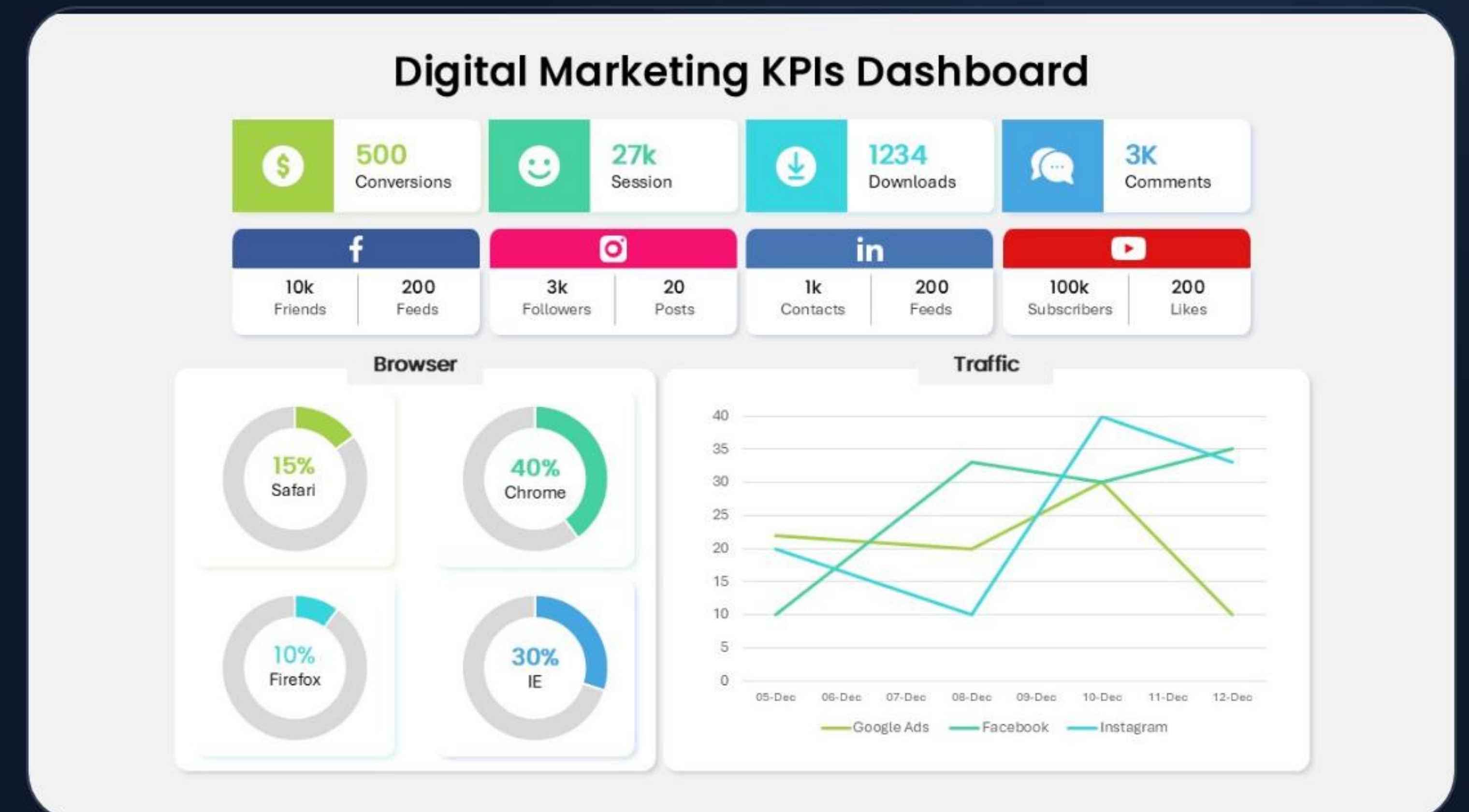
The optimizer enforces strict ****Skill Matching****: Junior resources are mathematically barred from Level 3 complexity tasks through extreme fitness penalties.

REAL-TIME PLANNING INTERFACE

Visibility & Control

Our dashboard provides transparency into the "Black Box" of optimization. Stakeholders can visualize:

- ✓ Task Start/End Timeline Estimates
- ✓ Developer Utilization Heatmaps
- ✓ Cost-Benefit Trade-off Analysis



DEPENDENCY AWARENESS

Directed Acyclic Graphs (DAGs)

Tasks are no longer treated as isolated blocks. The system honors "Finish-to-Start" relationships, ensuring the schedule respects the natural technical flow.

If Task B depends on Task A, the optimizer calculates the 'Ready Time' dynamically, preventing idle developer time and blocking bottlenecks.



FINANCIAL OPTIMIZATION RESULTS



~32% Estimated Reduction in Operational Spend

EVOLUTIONARY MECHANICS



True Elitism: Ensuring the #1 most efficient schedule in every generation is preserved and never corrupted.



Crossover & Mutation: Recombining "fit" schedules while maintaining genetic diversity to avoid local optima.



Makespan Focus: Penalizing long overall project durations to force parallel development.

Questions?

Thank you for exploring the future of AI-Driven Agile Management

Project Repository: [GitHub.com/YourRepo/Smart-Agile-Optimizer](https://github.com/YourRepo/Smart-Agile-Optimizer)
Python 3.10+ | Scikit-Learn | NumPy | Pandas

IMAGE SOURCES



<https://slidebazaar.com/wp-content/uploads/2025/06/digital-marketing-kpis-dashboard-powerpoint-google-slides.jpg>

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<https://www.arjankc.com.np/assets/images/posts/ma-integration-matrix-operations/featured.webp>

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